

Nevidu Jayatilleke

nevidu.25@cse.mrt.ac.lk | +94 71 251 4638 | Google Scholar | LinkedIn | Research Portfolio | GitHub

Profile

I am currently a Postgraduate Researcher at the University of Moratuwa, Sri Lanka, specialising in multilingual Natural Language Processing (NLP) with a dedicated focus on advancing AI capabilities for low-resource environments. My work is driven by the goal of bridging the digital language divide, ensuring that AI technologies are universally accessible and effective across diverse linguistic backgrounds. My primary research expertise lies in computational semantics and diachronic linguistics, specifically modeling the evolution of meaning across historical and canonical timeframes. My broader research interests include the development of diachronic corpora, machine translation for figures of speech, and benchmarking OCR performance for underrepresented scripts, alongside advancements in automatic text summarisation and information extraction. Through these academic pursuits, I strive to develop technologically inclusive systems capable of handling complex linguistic and cultural nuances on a global scale.

Education

University of Moratuwa, Sri Lanka , MSc (Major Research Component) – Full Time	Apr 2025 – Present
- Research Area: Diachronic Semantic Drift. - Supervised by Dr. Nisansa De Silva, Department of Computer Science and Engineering.	
Robert Gordon University, UK , BSc(Hons) Artificial Intelligence and Data Science	Sep 2020 – 2024 May
- First-class honours degree. Key Modules: Machine Learning, Simulations and Modelling Techniques, Computational Intelligence, Natural Language Processing, Advanced Mathematics for Data Science, Research Trends. - Research Project in the field of text summarisation of patent documents.	
Royal Institute International School, Sri Lanka	Sep 2014 – 2020 Jun
- CIE A Levels: 4As A Level Subjects: Mathematics, Further Mathematics, Physics, Computer Science	

Publications

SiDiaC-v. 2.0: Sinhala Diachronic Corpus Version 2.0	May 2026
<i>Nevidu Jayatilleke</i> , Nisansa De Silva, Uthpala Nimanthi, Gagani Kulathilaka, Azra Safrullah, Johan Sofalas <i>15th Language Resources and Evaluation Conference (LREC 2026)</i> 📄 #7 aclanthology.org/2026.lrec-1.537	
Dynamics of meaning: Towards the Evaluation of Diachronic Semantic Change in Sinhala	Accepted
<i>Nevidu Jayatilleke</i> , Nisansa De Silva <i>The 5th AACL & the 15th International Joint Conference on Natural Language Processing</i> 📄 #11 arxiv.org/abs/2609.08609	
SiDiaC: Sinhala Diachronic Corpus	Sep 2025
<i>Nevidu Jayatilleke</i> , Nisansa De Silva <i>39th Pacific Asia Conference on Language, Information and Computation (PACLIC 39)</i> aclanthology.org/2025.paclic-1.47	
Zero-shot OCR Accuracy of Low-Resourced Languages: A Comparative Analysis on Sinhala and Tamil	Sep 2025
<i>Nevidu Jayatilleke</i> , Nisansa De Silva <i>2025 International Conference on Recent Advances in Natural Language Processing</i> aclanthology.org/2025.ranlp-1.56	

A Hybrid Architecture with Efficient Fine Tuning for Abstractive Patent Document Summarization Nevidu Jayatilleke , Ruwan Weerasinghe <i>8th International Conference on Smart Computing and Systems Engineering, University of Kelaniya, Sri Lanka</i> 10.1109/SCSE65633.2025.11030964	Apr 2025
Advancements in Natural Language Processing for Text Summarization Nevidu Jayatilleke , Ruwan Weerasinghe, Nipuna Senanayake <i>4th International Conference on Computer Systems 2024 in Hangzhou, China,</i> 10.1109/ICCS62594.2024.10795848	Sep 2024
SinhaLegal: A Benchmark Corpus for Information Extraction and Analysis in Sinhala Legislative Texts Minduli Lasandi, Nevidu Jayatilleke <i>Second Workshop on Language Models for Low-Resource Languages (LoResLM 2026) @ EACL 2026</i> aclanthology.org/2026.loreslm-1.11	Mar 2026
SinFoS: A Parallel Dataset for Translating Sinhala Figures of Speech Johan Sofalas, Dilushri Pavithra, Nevidu Jayatilleke , Ruwan Weerasinghe <i>22nd Workshop on Multiword Expressions (MWE 2026) @ EACL 2026</i> aclanthology.org/2026.mwe-1.2	Mar 2026
SiPaKosa: A Comprehensive Corpus of Canonical and Classical Buddhist Texts in Sinhala and Pali Ranidu Gurusinghe, Nevidu Jayatilleke <i>2nd Workshop on Challenges in Processing South Asian Languages (CHiPSAL) @ LREC 2026</i> aclanthology.org/2026.chipsal-1.8	May 2026
From Sinhala to Dhivehi: Cross-Lingual Transfer Learning for Low-Resource Speech Recognition  Lukmal Ilyas, Nevidu Jayatilleke <i>12th International Moratuwa Engineering Research Conference (MERCon) 2026</i> doi.org/10.1109/MERCon71835.2026.11691501	Aug 2026
Cross-Temporal Sinhala OCR: Page-Level Adaptation and Diachronic Analysis Avisha Dilhara, Nevidu Jayatilleke <i>12th International Moratuwa Engineering Research Conference (MERCon) 2026</i> doi.org/10.1109/MERCon71835.2026.11691307	Aug 2026
LMSpell: Spell Correction with Pre-Trained Language Models Akeshe Gunathilake, Nadil Karunarathna, Tharusha Bandaranayake, Surangika Ranathunga, Nisansa de Silva, Nevidu Jayatilleke <i>12th International Moratuwa Engineering Research Conference (MERCon) 2026</i> doi.org/10.1109/MERCon71835.2026.11691371	Aug 2026
Impact of Cyclone Ditwah on the Colombo Stock Exchange: An Event Study on Market Reactions to Natural Disasters Anupama Wickramaratne, Thevindu Fernando, Tharupahan Jayawardana, Dehan Wijesinghe, Senindu Dinapura, Madara Mendis, Nevidu Jayatilleke , Nisansa de Silva, Sandareka Wickramanayake <i>12th International Moratuwa Engineering Research Conference (MERCon) 2026</i> doi.org/10.1109/MERCon71835.2026.11691317	Aug 2026
PlurVA-LLM-2026 Shared Task Track-1: Pluralistic Value Alignment in LLMs via Multilingual Fine-Tuning and Threshold Calibration Vihindi Kotalawala, Nevidu Jayatilleke	Accepted

SinBrief: A Hybrid Framework for Abstractive Text Summarisation of Sinhala Legal Documents

Accepted

Minduli Lasandi, *Nevidu Jayatilleke*

13th Conference on Computational Linguistics and Speech Processing (ROCLING) 2026

arxiv.org/abs/2609.32397

Experience

Teaching Assistant, University of Moratuwa Nov 2025 – Present

- Assessment marking for the *Advanced Machine Learning* and *Big Data Analytics* module applies to both Bachelor's and Master's students.
- Overseeing the Final Year Teaching Assistants to ensure smooth conduct of *Data Structures and Algorithms* module labs.
- Guide CSE undergraduate students in the Introduction to Data Science and Advanced NLP modules on research studies and paper writing.

Visiting Lecturer, Informatics Institute of Technology Nov 2025 – Present

- Supervising final year research projects (2026/2027) on building tokenisers for low-resource languages and word sense disambiguation.
- Conducted tutorial sessions on the modules; *Natural Language Processing*, *Final Year Project*, and *Software Development Fundamentals*.
- Supervised final year research projects (2025/2026) on Text Summarisation, QnA, Hallucination Detection, OCR, Paraphrasing, Machine Translation and ASR.

Research Assistant (Assistant Lecturer Grade), Informatics Institute of Technology Nov 2024 – Nov 2025

- Assisted in drafting the Sinhala component for a collaborative DFG grant proposal between IIT, University of Jaffna, and FAU Germany, focusing on low-resource LLM data scarcity, dialectal variation, and cultural awareness.
- Mentoring final-year students at IIT and UCSC as an advisor in research problems related to NLP and BioInformatics.
- Conduct lectures for students on Machine Learning, Deep Learning, NLP, and Research Trends.

Data Scientist, LeverEdge Sports – Sydney, Australia June 2024 – Jan 2025

- Developing a time series regression model using ensemble machine learning algorithms and deep learning architectures to predict a baseball team's (MLB) scores at different inning intervals during a game.
- Implementing Time Series Classification models to classify game-winner at different intervals of the selected sport game using FFNN and 1D-CNN architectures.
- Analysing NBA games to predict the winner at each quarter of ongoing games using statistical analysis methods.

Machine Learning Engineer Intern, IronOne Technologies Jul 2022 – Jul 2023

- Contributed to a financial fraud detection project using machine learning and deep learning architectures.
- Collaborated with domain experts on data imputation, augmentation, dimensionality reduction, encoding, and scaling.
- Leveraged Amazon Snowflake and SageMaker for data extraction, analysis, preprocessing, and modelling.
- Automated data preprocessing and notebooks to streamline pipelines and support MLOps.

Projects

Sinhala & Tamil Mini GPT Implementations 🔄

- PyTorch implementations of decoder-only Transformers (Atto-GPT, Pico-GPT) for character-level language generation in Sinhala and Tamil.

Sinhala Constitution RAG Chatbot 🔄

- Naive RAG architecture for the Sinhala Constitution, utilising specialised contextualised and non-contextualised

embedding techniques for the Sinhala language.

MLB & NBA Score Predictors 🧠🧠

- Time-series regression models, such as Feedforward Neural Networks (FFNN), 1D Convolutional Neural Networks (1D-CNN), and Ensemble methods, are used to predict final game scores.

MLB Winner Predictors 🧠

- Time-series classification models, including Feedforward Neural Networks (FFNN), 1D Convolutional Neural Networks (1D-CNN), and ensemble methods, are used to predict the final outcomes of games at various intervals.

Sinhala Text Preprocessing Resources 🧠

- Comprehensive suite of preprocessing tools, including suffix removal, stem dictionaries, tokeniser resources, and Word2Vec embeddings for Sinhala.

Technologies

Languages: Python, Java, R, SQL

Technologies: Tensorflow, Keras, Pytorch, Pandas, Langchain, Huggingface, Pandera, Prefect, Apache Airflow, Google Cloud Platform, AWS Sagemaker, Snowflake, GitHub, Docker

Achievements and Certificates

- IELTS with overall score 7.5:
 - Speaking 7.5
 - Writing 6.5
 - Reading 8.5
 - Listening 8.0
- Informatics Institute of Technology Senate Award for Research Excellence 2025.
- Successful completion of a workshop session on "Demystifying the Transformer: Build the GPT Core from Scratch" at the flagship ICTer 2025 conference organised by the University of Colombo, School of Computing.
- Best Paper Award (NLP Track), MERCon 2026: "From Sinhala to Dhivehi: Cross-Lingual Transfer Learning for Low-Resource Speech Recognition"

Reference

Dr. Nisansa de Silva

Senior Lecturer at University of Moratuwa
Email - NisansaDds@cse.mrt.ac.lk

Dr. Ruwan Weerasinghe

Dean - Research at Informatics Institute of Technology
Email - ruwan.w@iit.ac.lk